

TECHNICAL DOCUMENTATION

FortiGate High Availability

Active-Passive Cluster — Configuration, Priority Management & Failover Testing

Field	Details
Document Title	FortiGate HA Active-Passive Cluster — Configuration & Failover Guide
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1. Introduction & Purpose

High Availability (HA) is a core resiliency requirement for any network edge device. A single FortiGate firewall represents a single point of failure: if it goes offline due to a hardware fault, a software crash, or scheduled maintenance, every service that depends on it — internet access, VPN tunnels, and inter-site connectivity — goes down with it.

This document describes the design and configuration of a 4-unit FortiGate cluster running in Active-Passive mode using the Fortinet Clustering Protocol (FGCP). The goal is to guarantee that if the currently active unit fails, one of the remaining passive units automatically and transparently takes over, with minimal to zero disruption to ongoing traffic.

The document is organized to mirror the actual build sequence used in the lab: switch preparation, topology, per-unit configuration, HA profile setup, failover validation, and finally controlled priority-based failover — where a specific unit is deliberately forced to take over the Primary role.

2. Network Topology

The lab consists of four FortiGate units, each acting as the edge firewall for one site, all forming a single FGCP HA cluster. Each FortiGate connects to three types of neighbors:

- port1 — connects upward to the shared "Net" cloud (representing the WAN / upstream network).
- port4 — connects to a dedicated local client (VPC) representing that site's LAN.
- port2 and port3 — connect down to a central Layer 2 Switch. These two links form the redundant HA heartbeat path between all four cluster members.

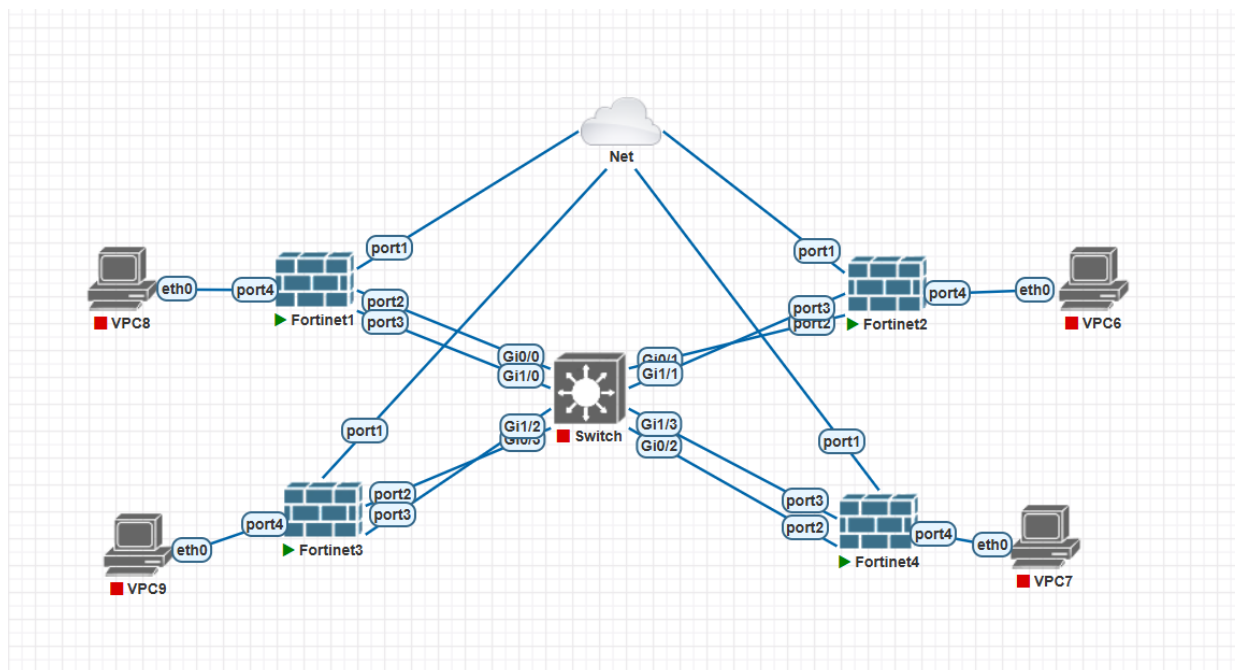


Figure 1 — Full lab topology: Net cloud, 4 FortiGate units, central Switch, and 4 client VPCs.

Device & Interface Mapping

FortiGate	Hostname	port1 (WAN → Net)	port4 (LAN → Client)	port2 / port3 (→ Switch)
Fortinet1	HQ1	Connected to Net	VPC8	HA Heartbeat
Fortinet2	HQ2	Connected to Net	VPC6	HA Heartbeat
Fortinet3	HQ3	Connected to Net	VPC9	HA Heartbeat
Fortinet4	HQ4	Connected to Net	VPC7	HA Heartbeat

3. Switch Configuration

Before touching the FortiGate units, the switch must be prepared to carry HA heartbeat traffic between all four members. Since port2 and port3 of every FortiGate land on this switch, all eight of those switchports are placed into a single, dedicated VLAN — isolating heartbeat traffic from any other network segment and ensuring the four units can reach each other at Layer 2.

3.1 Create the Dedicated HA VLAN

A VLAN is created and named HA so its purpose is unambiguous to anyone reviewing the switch configuration later.

```
enable
configure terminal

vlan 10
  name HA
exit
```

3.2 Assign the Heartbeat Ports to the VLAN

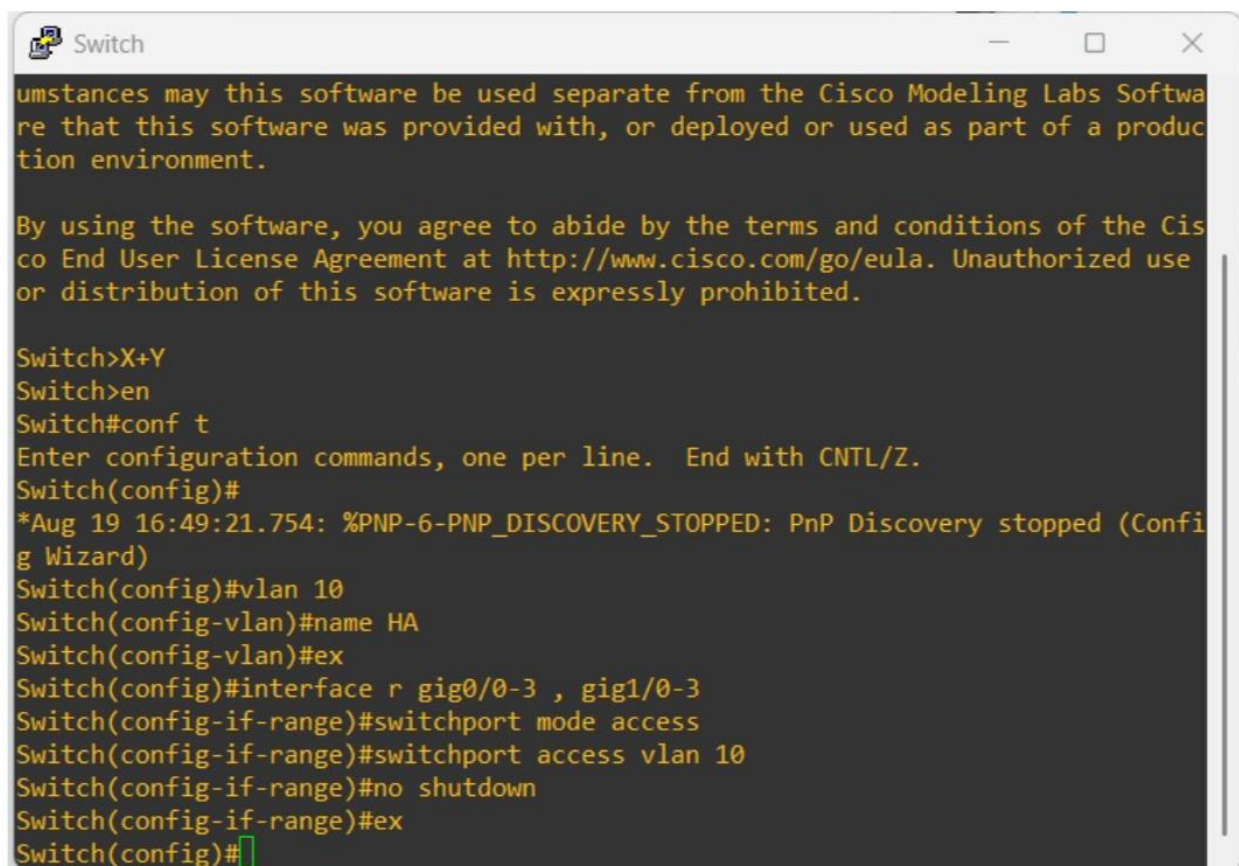
Every switchport connected to a port2 or port3 of any FortiGate (Gi0/0, Gi1/0, Gi0/1, Gi1/1, Gi1/2, Gi0/3, Gi1/3, Gi0/2 in this lab) is set as an access port in VLAN 10.

```
interface range GigabitEthernet0/0 - 3 , GigabitEthernet1/0 - 3
  switchport mode access
  switchport access vlan 10
  no shutdown
exit

end
write memory
```

3.3 Verify the VLAN

```
show vlan brief
show interfaces status
```



```

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Switch>X+Y
Switch>en
Switch#conf t
Enter configuration commands, one per line. End with CNTL/Z.
Switch(config)#
*Aug 19 16:49:21.754: %PNP-6-PNP_DISCOVERY_STOPPED: PnP Discovery stopped (Config Wizard)
Switch(config)#vlan 10
Switch(config-vlan)#name HA
Switch(config-vlan)#ex
Switch(config)#interface range gig0/0-3 , gig1/0-3
Switch(config-if-range)#switchport mode access
Switch(config-if-range)#switchport access vlan 10
Switch(config-if-range)#no shutdown
Switch(config-if-range)#ex
Switch(config)#

```

4. FortiGate Base Configuration (Per Unit)

The following steps are repeated identically on all four FortiGate units before HA is enabled. Only the hostname and IP values differ per unit, as shown in the table below.

4.1 Set the Hostname

Navigate to System > Settings and set the Host Name field for each unit:

Device	Hostname to Set
Fortinet1	HQ1
Fortinet2	HQ2
Fortinet3	HQ3
Fortinet4	HQ4

4.2 Configure port1 (WAN, facing Net)

1. Go to Network > Interfaces and edit port1.

2. Set Role to WAN.
3. Set Addressing mode to Manual and assign an IP/Netmask reachable by the Net cloud.
4. Under Access, enable PING (and HTTPS/SSH only if this port is also used for remote management).
5. Click OK.

4.3 Configure port4 (LAN, facing the local VPC)

1. Edit port4 and set Role to LAN.
2. Set Addressing mode to Manual and assign an IP/Netmask for that site's local subnet.
3. Click OK.

port2 and port3 are intentionally left unconfigured at this stage — they will be automatically reserved as HA heartbeat interfaces once HA mode is enabled in the next section.

5. HA Profile Configuration on FortiGate

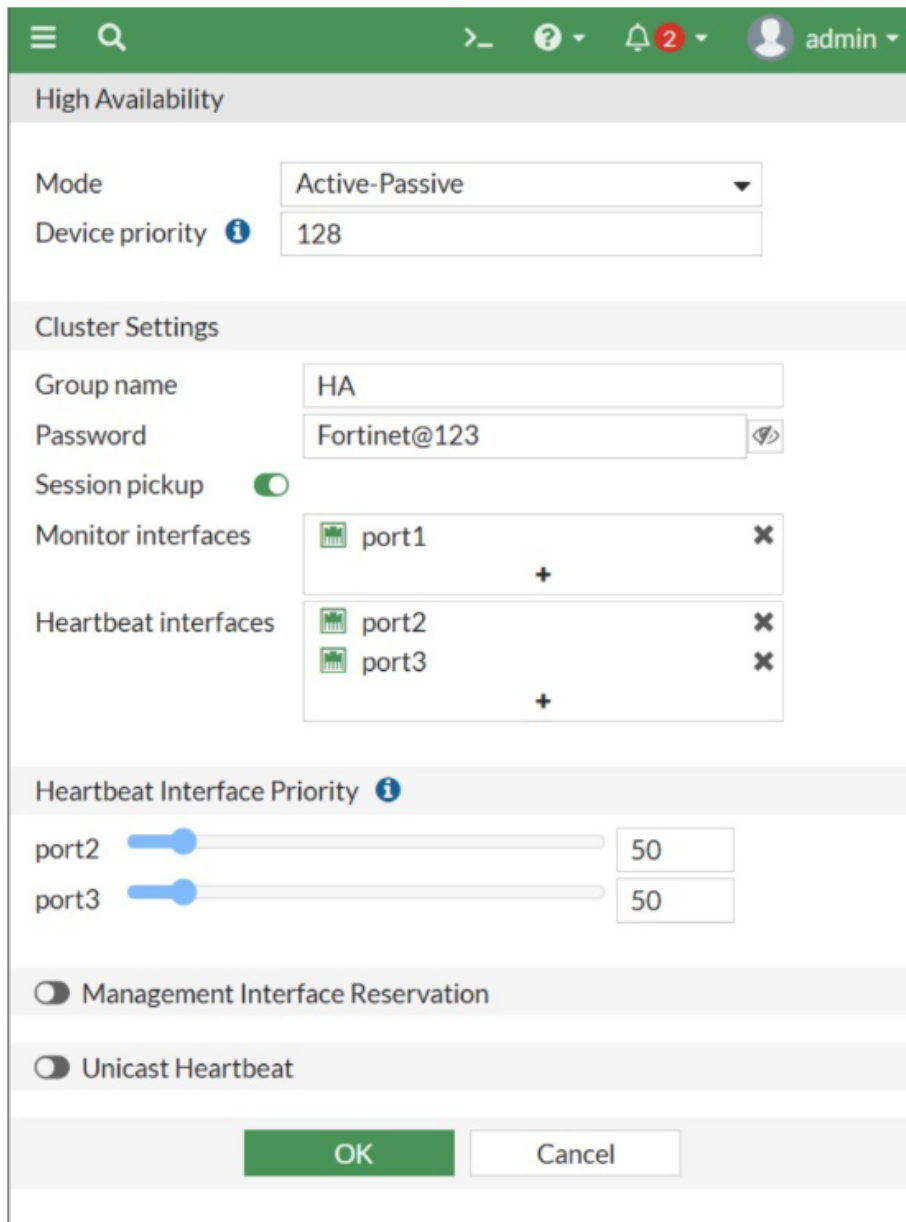
With base interfaces configured, HA is enabled identically on all four units via System > HA. This is what actually forms the cluster: as soon as two or more units share the same Group Name and Password on their heartbeat network, they discover each other and begin electing a Primary unit.

Field	Value (identical on all 4 units)
Mode	Active-Passive
Group Name	HA-CLUSTER
Password	Set a strong shared password
Heartbeat Interfaces	port2 and port3, both enabled
Session Pickup	Enabled — preserves active sessions across failover
Override	Disabled, except where a controlled failover explicitly requires it (see Section 8)

Device Priority (the only value that differs per unit)

Device Priority determines election order: the unit with the highest priority becomes Primary; if it fails, the unit with the next-highest priority takes over, and so on.

Hostname	Device Priority	Expected Role
HQ1	250	Primary (Active) under normal conditions
HQ2	200	First failover candidate
HQ3	150	Second failover candidate
HQ4	100	Last failover candidate



The image shows the FortiGate HA configuration interface. At the top is a green header bar with navigation icons and a user profile labeled 'admin'. Below the header is a 'High Availability' section with a 'Mode' dropdown set to 'Active-Passive' and a 'Device priority' field set to '128'. The 'Cluster Settings' section includes a 'Group name' field set to 'HA', a 'Password' field set to 'Fortinet@123', and a 'Session pickup' toggle that is turned on. Below these are two lists of interfaces: 'Monitor interfaces' with 'port1' and 'Heartbeat interfaces' with 'port2' and 'port3'. Each interface entry has a small icon, a plus sign, and a delete 'X' icon. The 'Heartbeat Interface Priority' section shows sliders for 'port2' and 'port3', both set to '50'. At the bottom are two unchecked options: 'Management Interface Reservation' and 'Unicast Heartbeat'. The interface concludes with 'OK' and 'Cancel' buttons.

High Availability

Mode: Active-Passive

Device priority: 128

Cluster Settings

Group name: HA

Password: Fortinet@123

Session pickup: ☒

Monitor interfaces: port1

Heartbeat interfaces: port2, port3

Heartbeat Interface Priority

port2: 50

port3: 50

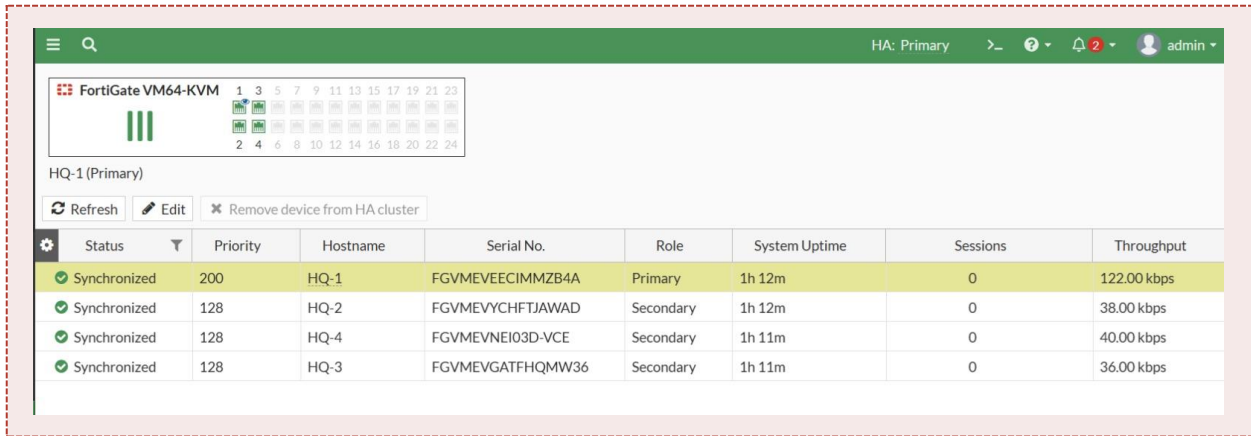
☐ Management Interface Reservation

☐ Unicast Heartbeat

OK Cancel

Cluster Verification

After enabling HA on all four units, allow a short negotiation period, then confirm the cluster has formed correctly.



The screenshot shows the FortiGate HA Active-Passive Cluster GUI. At the top, it indicates 'HA: Primary' and the user 'admin'. Below this, there's a status bar for 'FortiGate VM64-KVM' showing a grid of 24 units. The main table lists the units in the cluster:

Status	Priority	Hostname	Serial No.	Role	System Uptime	Sessions	Throughput
✓ Synchronized	200	HQ-1	FGVMEVEECIMMZB4A	Primary	1h 12m	0	122.00 kbps
✓ Synchronized	128	HQ-2	FGVMEVYCHFTJAWAD	Secondary	1h 12m	0	38.00 kbps
✓ Synchronized	128	HQ-4	FGVMEVNEI03D-VCE	Secondary	1h 11m	0	40.00 kbps
✓ Synchronized	128	HQ-3	FGVMEVGATFHQMW36	Secondary	1h 11m	0	36.00 kbps

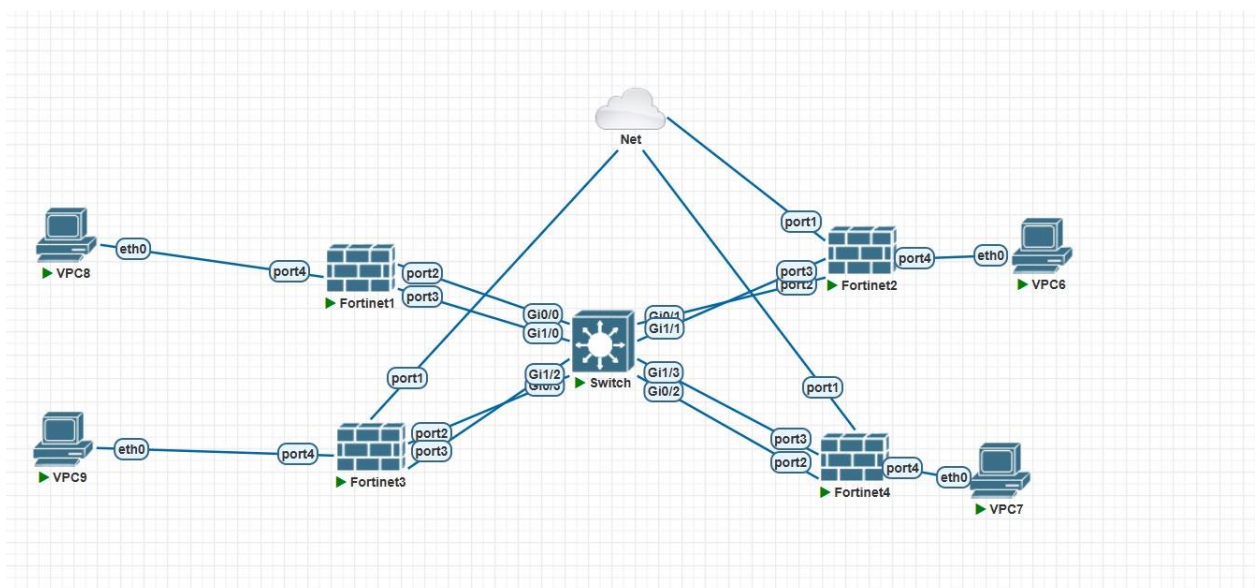
6. Failover Testing

Two independent failover scenarios are used to validate that the cluster behaves correctly under different types of failure — a full unit outage, and a single-link outage on the active unit.

Failover Test 1 — Full Unit Failure

This test simulates a complete outage of the current Primary unit (power loss, hardware failure, or reboot).

1. Confirm the current Primary unit from System > HA (expected: HQ1).
2. Start continuous traffic from a client (e.g., ping from VPC8 through the Net).
3. Power off / disconnect the current Primary unit entirely.
4. Observe the remaining units renegotiate and elect a new Primary based on Device Priority.
5. Confirm on the client side that traffic resumes automatically after a brief interruption.

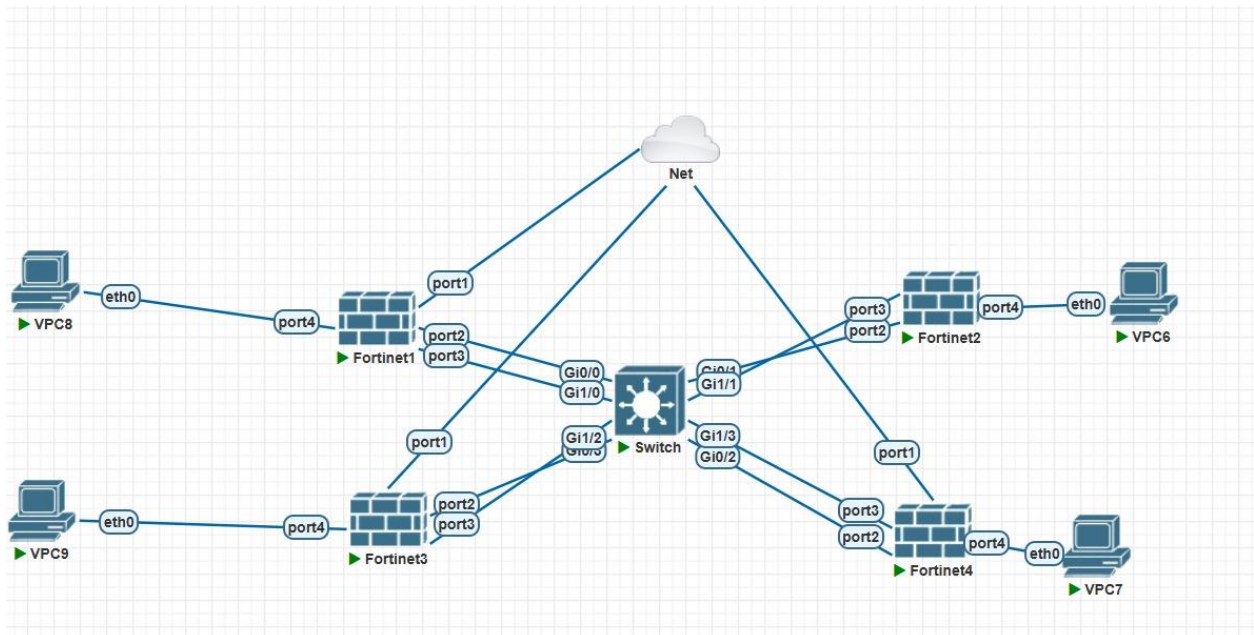


FortiGate VM64-KVM 1 3 5 7 9 11 13 15 17 19 21 23 2 4 6 8 10 12 14 16 18 20 22 24 HQ-2 (Primary) Refresh Edit Remove device from HA cluster							
Status	Priority	Hostname	Serial No.	Role	System Uptime	Sessions	Throughput
✓ Synchronized	128	HQ-2	FGVMEVYCHFTJAWAD	Primary	1h 18m	0	103.00 kbps
✓ Synchronized	128	HQ-4	FGVMEVNEIO3D-VCE	Secondary	1h 18m	0	229.00 kbps
✓ Synchronized	128	HQ-3	FGVMEVGATFHQMW36	Secondary	1h 18m	0	39.00 kbps
✓ Synchronized	128	HQ-1	FGVMEVEECIMMZB4A	Secondary	1h 18m	0	241.00 kbps

Failover Test 2 — Link (Interface) Failure

This test simulates a partial failure: the FortiGate itself stays powered on, but its WAN link goes down. This validates that HA reacts to link-level failures, not just full device outages (Link Monitoring / Remote Link Failover).

1. Confirm the current Primary unit from System > HA.
2. Start continuous traffic from a client through the Net, same as in Test 1.
3. Physically disconnect (or shut down) only port1 (WAN) on the current Primary unit, leaving the unit itself powered and reachable on the heartbeat network.
4. Observe that the cluster detects the WAN link loss and fails traffic over to the next-priority unit, even though the original Primary is still technically "alive" on the heartbeat link.
5. Reconnect port1 on the original unit afterward and confirm whether it rejoins as Secondary (expected, since Override is disabled).



HA: Primary

?

2

admin

FortiGate VM64-KVM

1 3 5 7 9 11 13 15 17 19 21 23

2 4 6 8 10 12 14 16 18 20 22 24

HQ-4 (Primary)

Refresh

Edit

Remove device from HA cluster

Status	Priority	Hostname	Serial No.	Role	System Uptime	Sessions	Throughput
✓ Synchronized	128	HQ-4	FGVMEVNEI03D-VCE	Primary	1h 19m	0	275.00 kbps
✓ Synchronized	128	HQ-2	FGVMEVYCHFTJAWAD	Secondary	1h 19m	0	53.00 kbps
✓ Synchronized	128	HQ-3	FGVMEVGATFHQMW36	Secondary	1h 19m	0	53.00 kbps
✓ Synchronized	128	HQ-1	FGVMEVEECIMMZB4A	Secondary	1h 19m	0	56.00 kbps

7. Priority Management & Controlled Failover

Beyond reacting to unplanned failures, Device Priority can be used to deliberately control which unit takes over the Primary role — for example, during planned maintenance on the current Primary. This section demonstrates a controlled failover where HQ2 is forced to become the new Primary as soon as HQ1 goes offline, purely as a result of HQ2 holding the next-highest Device Priority (200).

How Priority-Based Election Works

- Every cluster member advertises its Device Priority over the heartbeat link.
- When the current Primary becomes unreachable, the remaining members compare priorities.
- The member with the highest remaining priority — in this design, HQ2 at 200 — is elected the new Primary.
- Because Override is disabled, this election is based purely on priority and uptime, not on which unit was Primary previously — giving predictable, repeatable failover behavior.

Controlled Failover Procedure

1. Confirm current cluster state: HQ1 = Primary (250), HQ2 = Secondary (200), HQ3 = Secondary (150), HQ4 = Secondary (100).
2. Take HQ1 out of service in a controlled manner (graceful shutdown from the GUI, or Maintenance/Reboot), rather than a hard power failure.
3. Monitor System > HA from HQ2, HQ3, or HQ4 during the transition.
4. Confirm that HQ2 — and specifically HQ2, not HQ3 or HQ4 — is elected the new Primary, validating that priority 200 correctly outranks 150 and 100.
5. Confirm client traffic continues to flow with minimal interruption throughout the transition.
6. When HQ1 is brought back online, confirm it rejoins the cluster as a Secondary unit rather than reclaiming the Primary role automatically.

The top screenshot shows the 'HA Peer Configuration' page. It has a green header with a search icon, a menu icon, and the text 'HA: Primary'. Below the header, there are input fields for 'Peer' (set to 'HQ-2') and 'Priority' (set to '200'). A yellow warning box states: 'You will be logged out if a change in priority causes the primary HA device to be renegotiated.' At the bottom are 'OK' and 'Cancel' buttons.

The bottom screenshot shows the 'HA Status' page. It has a green header with a search icon, a menu icon, and the text 'HA: Primary'. Below the header, there is a 'FortiGate VM64-KVM' section with a grid of 24 icons. Below this is a table titled 'HQ-2 (Primary)' with buttons for 'Refresh', 'Edit', and 'Remove device from HA cluster'. The table has 8 columns: Status, Priority, Hostname, Serial No., Role, System Uptime, Sessions, and Throughput.

Status	Priority	Hostname	Serial No.	Role	System Uptime	Sessions	Throughput
✓ Synchronized	200	HQ-2	FGVMEVYCHFTJAWAD	Primary	1h 30m	0	75.00 kbps
✓ Synchronized	128	HQ-1	FGVMEVEECIMMZB4A	Secondary	1h 30m	0	46.00 kbps
✓ Synchronized	128	HQ-3	FGVMEVGATFHQMW36	Secondary	1h 30m	0	45.00 kbps
✓ Synchronized	128	HQ-4	FGVMEVNEI03D-VCE	Secondary	1h 30m	0	33.00 kbps

8. Troubleshooting Tips

- Cluster not forming: verify Group Name and Password are identical, character-for-character, on all four units.
- Heartbeat not coming up: confirm all eight switchports (port2/port3 uplinks) are correctly assigned to VLAN 10 (HA) and are in an 'up' state.
- Wrong unit becomes Primary: double-check the Device Priority value on each unit — a typo here is the most common cause of unexpected election results.
- Unit rejoins as Primary unexpectedly after recovery: check whether Override is enabled somewhere in the cluster — it should remain disabled for predictable priority-based behavior.
- Slow failover: review the heartbeat interval and lost-heartbeat threshold under the HA settings; lowering these values speeds up detection at the cost of more heartbeat traffic.

9. Conclusion

This configuration results in a 4-unit FortiGate Active-Passive cluster with a dedicated, VLAN-isolated heartbeat path, validated behavior under both full unit failure and single-link failure, and a predictable, priority-driven election process that has been explicitly proven by forcing HQ2 into the Primary role.

Once populated with the requested screenshots, this document is ready to serve as both a deployment record and a reference runbook for future maintenance or failover events.